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BMI 500 - Homework 7

HW1 Hypothesis:

(i) We know $S(0)=1995$, $I(0)=5$, $R(0)=0$, and transmission rate $\beta=0.125\times10^{-3}$, and recovery $\gamma=0.1$

simulate the Euler approximation with 5-days steps:

$$S_{t+5}=S_t-5\beta SI$$

$$I_{t+5}=I_t+5(\beta SI-\gamma I)$$

$$R_{t+5}=R_t+5\gamma I_t$$

e.g. $S_5=1995-5(0.125\times10^{-3}\times1995\times5)$

≈ 1988.77

$I_5=5+5(0.125\times10^{-3}\times1995\times5-0.1\times5)$

≈ 8.73

$R_5=0+5(0.1\times5)=2.5$

Day	Susceptible	Infections
0	1995	5
5	1989	9
⋮	⋮	⋮
45	1139	443 ↑
50	824	536
55	548	545 ↓
60	361	459

Initially, the new infection rate is: $\beta SI=0.125\times10^{-3}\times1995\times5\approx1.247$

the recovery rate is: $\gamma I=0.1\times5=0.5$

Thus, infections initially increased by: $1.247-0.5=0.747$ people/day

The peak occurs when new infections equal recoveries: $\beta SI=\gamma I$

$$S=\frac{\gamma I}{\beta I}=\frac{0.1}{0.125\times10^{-3}}=800$$

Based on the ~~5 steps~~ 5-days step simulation, 800 occurs around day 50 with $S\approx824$ and $I\approx536$.

I predict that the infectious populations will peak around day 50 with approx 536 infected people.

(II) I expect multiple waves over 1500 days, with decreasing peak size.

Based on $R_0=\frac{\beta N\delta}{(\delta+\mu)(\gamma+\mu)}\approx2.33$. Since $R_0>1$, I expect an initial outbreak. As susceptible individuals become infected, the effective reproduction number falls. However, births continuously replenish susceptibles at $\mu N=10$ ppl/day. This may allow renewed transmission after the initial outbreak. I therefore expect additional small waves.

(III) I predict that trasmission rate β will have a slightly greater effect on the peak infectious than recovery rate γ.

Based on part D, increasing β by 5 times will increase R_0 by 5 times, whereas increase γ by 5 times will ~~increase~~ decrease R_0 by $(0.25+0.005)/(0.05+0.005)\approx4.64$.

These effects are comparable, but β has a slightly larger effect. Since R_0 does not directly determine the infection peak, ~~we~~ this prediction must be tested ~~by~~ with the simulation.